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Erratum: Fullshape power spectrum for the Symmetron modified gravity model

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This erratum corrects the results presented in ref. [1], which were affected by an implementation error in the code. In particular, two issues were identified: (i) the parameter m_0 was implemented with inconsistent units, and (ii) the prior adopted for the parameter β_0 was overly restrictive, leading to an overestimation of its inferred value. In the corrected analysis, m_0 is consistently expressed in units of $h\,\mathrm{Mpc}^{-1}$, while the prior on β_0 is taken to be $\mathcal{U}(-1.0, 1.0)$; in addition, we have promoted the nuisance parameter $\tilde{c}$ to a free parameter with prior $\mathcal{U}(-1.0, 1.0)$ and the rest of the priors are as in reference [1]. The fiducial cosmology is $\{\Omega_{cdm}, \Omega_b, h, \log(10^{10}A_s), n_s\} = \{0.31, 0.049, 0.67, 3.066, 0.97\}$. We now present the corrected MCMC analysis. As a result, figures 8 and 12 of ref. [1], together with the corresponding tables reporting the best-fit parameter values, have been updated. The corrected version of figure 8 is shown here as figure 1 and is accompanied by table 1. Similarly, the corrected version of figure 12 is shown as figure 2 and is accompanied by table 2. The corresponding changes have also been incorporated into the updated arXiv version of the paper (https://arxiv.org/pdf/2506.09304). Figure 1 shows the posterior contours obtained from the parameter estimation for the Symmetron model using the EZMock data. The corresponding parameter constraints for the Symmetron model are reported in table 1. The purpose of this analysis is to verify that the correct GR behavior is recovered when the modified-gravity parameter β_0 is allowed to vary. Similarly, corrected results for the Hu-Sacwicki $n = 1$ $f(R)$ model are presented in figure 2 and table 2.

The updated results provide a good fit to the data and recover the expected ΛCDM behavior, indicated by the dashed lines in figure 1 and figure 2.

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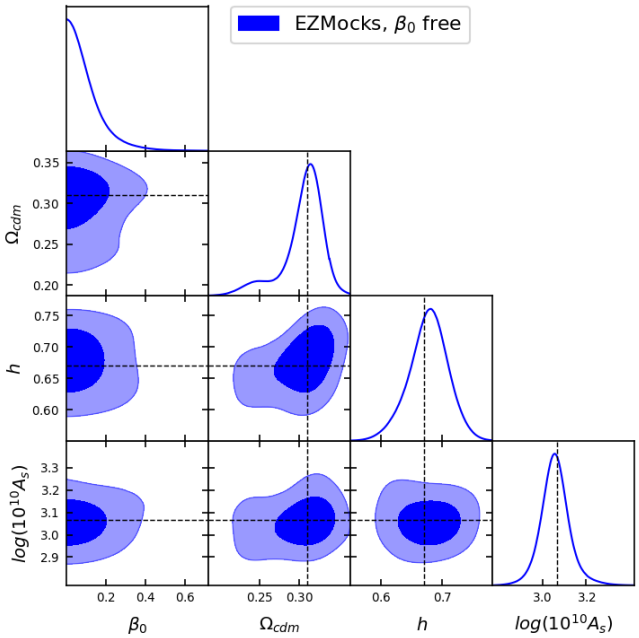


Figure 1. Confidence level contours for the parameter estimation of the EZMocks at $z = 0.75$ for Symmetron model. Using monopole and quadrupole data, our results show that we recover the simulation cosmological values employed in the EZMocks that are marked as dashed lines.

	EZMocks, free β_0
β_0	$< 2.8 \times 10^{-1}$
Ω_m	$0.313^{+0.009}_{-0.025}$
h	$0.679^{+0.027}_{-0.032}$
$\log(10^{10}A_s)$	$3.05^{+0.052}_{-0.053}$
b_1	$1.698^{+0.281}_{-0.280}$

Table 1. Marginalized constraints for the MG parameter (β_0) at 95% confidence level and the rest of the parameters at 68% confidence level.

	EZMocks, free f_{R0}
$\lvert f_{R0}\rvert$	$< 3 \times 10^{-4}$
Ω_m	$0.313^{+0.008}_{-0.028}$
h	$0.681^{+0.023}_{-0.026}$
$\log(10^{10}A_s)$	$3.053^{+0.050}_{-0.049}$
b_1	$1.668^{+0.291}_{-0.265}$

Table 2. Marginalized constraints for the $|f_{R0}|$ parameter at 95% confidence level, cosmological parameters, and local bias b_1 at 68% confidence level.

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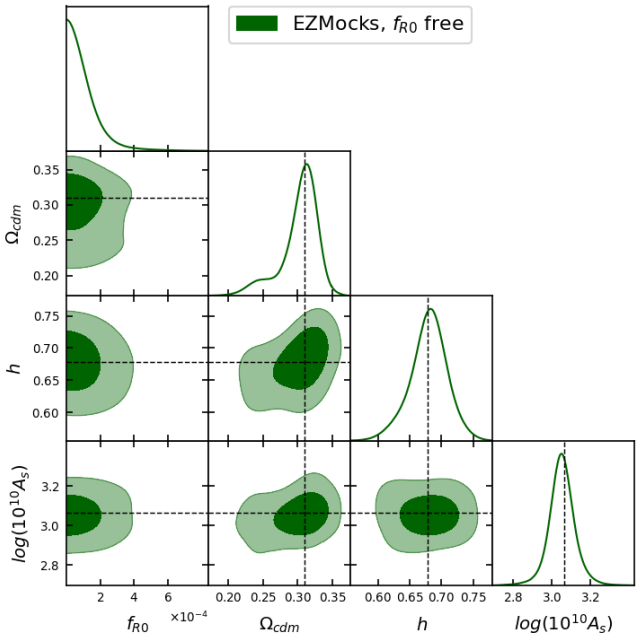


Figure 2. Confidence level contours for the parameter estimation performed on the EZMock realizations at $z = 0.75$. The analysis employs the monopole and quadrupole power spectrum multipoles, in which the parameter f_{R0} is left free in the fitting. The best-fit values obtained are summarized in table 2.

References

[1] G. Morales-Navarrete and J.L. Cervantes-Cota, *Fullshape power spectrum for the Symmetron modified gravity model*, *JCAP* **03** (2026) 079 [arXiv:2506.09304] [INSPIRE].

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